PHY-275 Fall 2026 Assignment #8 due 11/9/26 9:00pm

You will receive 1 credit **per task**.

**Task #1: Getting intuition with the $\delta$ operator**: prove that it is very much like the differential operator $d$ that it is

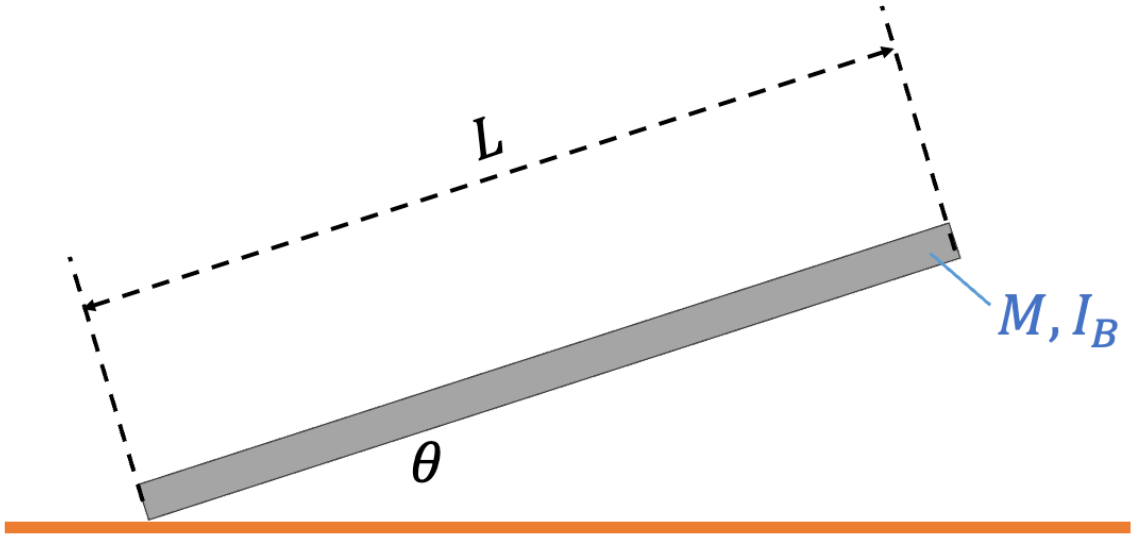
1) Commutative:

$$\delta\left(\frac{dy}{dx}\right) = \frac{d}{dx}(\delta y)$$

2) Follow the chain rule:

$$\delta(x^2) = 2x\delta x$$

**Task #2: Lagragian mech of a sliding rod (assuming no friction):**




Drive the kinetic energy two ways:

1):

$$T = \frac{1}{2}m\dot{x}^2 + \frac{1}{2}m\dot{y}^2$$

2*):*

$T = T_{trans} + T_{rot}$ *(like what Sam did in class)*

Did you get the same conclusion?

Then finish the rest of the work and derive the equation of motion.